

On the detection, modeling, and mitigation of anomalous QoT fluctuations in optical networks

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Monitoring data from deployed backbone networks shows that many lightpaths present large fluctuations of their quality of transmission (QoT), beyond 1 dB in range. In this paper, we investigate the hypothesis that the evidenced fluctuations stem from anomalies in polarization-dependent loss (PDL), equivalent to one large PDL element. We eventually propose and experimentally test a closed-loop automation designed to stabilize the QoT at a maximal level for lightpaths impaired by such a PDL anomaly. © 2024 Optica Publishing Group

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1. INTRODUCTION

With the advent of generalized optical performance monitoring, quality of transmission (QoT) time series of lightpaths monitored over prolonged time periods are now available for analysis. These time series are proving to be extremely rich sources of information. Indeed, many service-impacting phenomena observed in deployed networks significantly deviate from typical expectations inferred from the literature [1]. This has been notably evidenced with the observations of reversed aging [2] and Q-drops [3–5] in several datasets, including a public one gathering 4000 QoT time series collected in a large optical backbone [6]. In this paper, as an extension of a previous work [7], we focus on large QoT fluctuations as another anomaly type.

Polarization-dependent loss (PDL) is widely known to induce QoT fluctuations in standard, polarization-multiplexed (PM) coherent communications [8,9]. These fluctuations result from the combination of optical devices exhibiting PDL and random variations of state of polarization (SOP) in between them [10,11]. The main known sources of PDL are the wavelength selective switches (WSSs) in routing nodes [12,13] and the optical amplifiers [14]. Other network elements may also exhibit PDL, for instance, fiber bends [15] or more generally any fiber portion undergoing non-centrosymmetric strain [16].

To model the impact of PDL on a lightpath's QoT, any device with PDL is treated as the concatenation of the PDL-free device and a pure PDL element [17] characterized by its magnitude β , the ratio of optical attenuation between its two principal axes in Jones space [18]. Typical PDL magnitudes range from 0.1 to 0.5 dB [19]. For a typical lightpath, the

respective QoTs of both orthogonally polarized tributaries H and V may deviate by up to 1 dB in extreme, low-probability cases [9]. Yet, commercial transponders interleave data from both tributaries in the FEC frame so that error correction does not depend on the worst pre-FEC BER but on the average value over both tributaries [20]. As a result, the compound QoT for the PM signal, e.g., the monitored Q^2 , has significantly lower fluctuations than that of the individual QoTs of the tributaries [18]. Thus, typical distributed PDL may hardly explain two main field observations evidenced in this paper: first, that Q^2 fluctuations can exceed 1 dB in range; second, that extreme values are relatively too frequent to result from distributed PDL, as in [9]. We further show that these observations are however consistent with the presence of PDL anomalies, i.e., defects resulting in abnormally large PDL magnitude, or equivalent PDL impairment. We further propose a closed-loop scheme for the automated mitigation of the impact of such PDL anomalies achieving stabilization of the QoT around its highest possible value.

In the following, we first investigate in Section 2 the statistical distribution of QoT fluctuations in a large optical backbone network, leveraging the public dataset accessible in [6]. To that aim, we introduce a numerical method that specifically measures fluctuation levels by removing interferences from other QoT anomalies. Then, we leverage in Section 3 an extended analytical model derived in Appendix A to investigate which equivalent PDL anomaly may explain the large QoT fluctuations observed in the dataset. In Section 4, we perform numerical simulations to study the impact of a single PDL anomaly within a realistic lightpath with distributed PDL, and the possibility to model these lightpaths with a single PDL

equivalent. Finally, in Section 5, we propose and experimentally test a method specifically designed to mitigate large QoT fluctuations induced by PDL anomalies.

2. DETECTION OF LARGE QOT FLUCTUATIONS IN LIVE OPTICAL NETWORKS

Large fluctuations of the QoT can represent a severe challenge for optical network operations. Indeed, if their impact is not mitigated at design by adding margins, QoT fluctuations may repeatedly bring the instantaneous QoT below the FEC limit, inducing temporary packet losses. Furthermore, in the elastic optical network paradigm [21], QoT instability results in more conservative approaches for rate adaptation, hindering the network's optimization potential [22]. In this section, we present a numerical method to automatically detect QoT fluctuations in monitoring datasets. We further apply the method to study the QoT fluctuations in field-collected data [6].

A. Detection Challenge

QoT fluctuations may be measured within a time window through a variety of well-known metrics [23] such as the standard deviation, the median absolute deviation (MAD), the inter-quantile range (IQR), or the range, i.e., the difference between the maximum and minimum values observed. Since QoT fluctuation levels may significantly evolve in time, we aim to measure fluctuations locally, i.e., around every sample through a sliding window of width w . The width w is ideally chosen larger than the typical period of the fluctuations.

In Fig. 1, we plot as a reference QoT metric the raw Q^2 [24] for the dataset's lightpath #2252. Note that the raw Q^2 values are provided with a 15-min refresh period, and are the averages of the instantaneous Q^2 over the last 15 min. Consequently, fast fluctuations of the instantaneous Q^2 , with characteristic times shorter than the 15-min averaging period, cannot be observed here. We thus focus on fluctuations of the average QoT over longer periods, considering moving standard deviation (MSD) and moving range with $w = 1$ day as plotted in Fig. 1.

Although the two metrics properly characterize the level of QoT fluctuations in most instances, we observe that a Q-drop

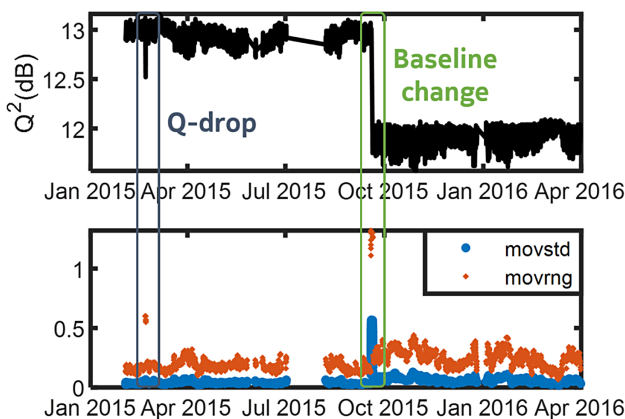


Fig. 1. Q-drop and QoT baseline change within the time series (top) pollute the moving metrics calculated on the raw data, here the standard deviation *movstd* and range *movrng*.

and a baseline change lead to significantly higher values. More specifically, Q-drops and other outliers tend to distort the measured range while baseline changes impact both standard deviation and range. Such anomalies are thus likely to generate false positives when attempting to detect large fluctuations. In the following, we present numerical methods to remove Q-drops and baseline changes from the QoT time series, allowing robust characterization of local fluctuation levels.

B. Detection and Removal of Q-Drops and Other Outliers

We leverage Tukey's method to detect Q^2 outliers [25]. We label as outlier any sample that is more than five scaled moving MAD away from the moving median. Note that a Q-drop is simply defined as a low Q^2 outlier, below the baseline. Then, each outlier sample is replaced by the average of the two neighbor samples. Outlier detection is illustrated for the previous time series in Fig. 2, displaying the upper and lower outlier detection thresholds.

With the tested method, the undetected Q-drops—the false negatives—are too small to significantly impact the local fluctuation levels. Regarding false positives, we observe, e.g., between February 23rd and 26th, that non-outlier samples may be very close to an outlier threshold, particularly the upper one here. As an inherent limitation of the detection method, some samples may be incorrectly labeled as outliers. However, since these false positives are replaced by an average of the closest non-outlier samples, they have a negligible impact on the measured fluctuation levels.

C. Baseline Correction

The frequent occurrences of sudden baseline changes constitute the major challenge for achieving automatic characterization of QoT fluctuations at network scale, as illustrated in Fig. 1. Thus, we leverage an algorithm proposed by Killick *et al.* that specifically detects such changes, i.e., changes of the mean, within time series [26]. Then, we calculate the piecewise QoT average in between detected changes and subtract this baseline from the data after outlier removal, thus isolating QoT

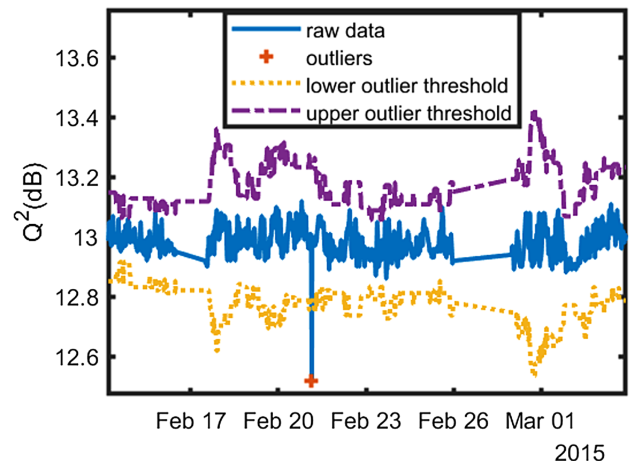


Fig. 2. Outlier detection with upper and lower outlier detection thresholds following the local QoT fluctuations.

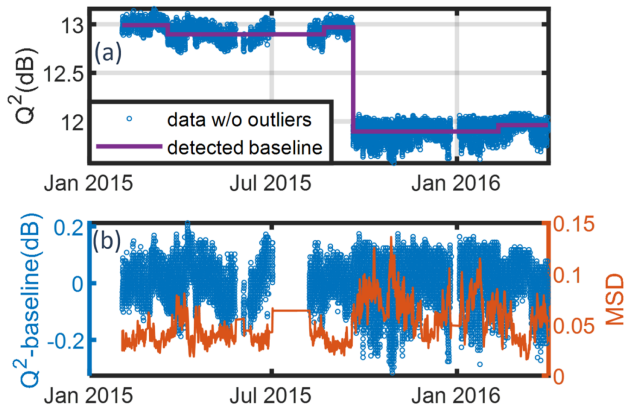


Fig. 3. (a) Detected baseline from the QoT after outlier removal and (b) isolated fluctuations after subtracting the baseline from the QoT, and corresponding MSD with 24-h moving window.

fluctuations. Finally, we measure the MSD of the processed data to measure local fluctuation levels.

The outcome of the baseline change detection is illustrated in Fig. 3(a), where the threshold K [26] is set at 10, leading to a medium sensitivity of the change detection. Note that K corresponds to a variance of Q^2 in dB. We observe that the algorithm automatically detects several changes in the time series, the most critical to our application being the 1 dB baseline change near October 1st. After baseline removal as illustrated in Fig. 3(b), we observe that the local QoT fluctuation evaluated by the MSD is not impacted by this baseline change, nor by the other minor ones detected within the time series. We further observe that the MSD successfully grasps the evolution of the fluctuation levels throughout the time series.

D. Quantifying QoT Fluctuations in an Optical Backbone

Following the numerical method described above, we evaluate the MSD with $w = 1$ day for the 4000 QoT time series of the dataset after outlier and baseline removal. We first consider the median MSD value for each lightpath to rank them by growing fluctuation level for the entire monitoring period. In Fig. 4, we show excerpts of the QoT time series for the lightpaths with the two largest median MSD values. For both, we observe fluctuations of varying speeds and ranges exceeding 1 dB. We also observe from Figs. 4(a.1), 4(a.2), 4(b.1), and 4(b.2) that the QoT for both lightpaths is not distributed as it would if the fluctuations were due to distributed PDL [9]. Alternatively, observed QoT fluctuations could be related to optical power fluctuations, due to instabilities from one or more of the inline amplifiers [27]. However, we checked that for both lightpaths represented in Fig. 4, the measured received power, also provided in the dataset, remained very stable, within ± 0.01 dBm, and was not correlated at all with the Q^2 . Thus, the observed fluctuations are more consistent with a single PDL element of high magnitude and randomly varying orientation [18]. We further investigate this in Sections 3 and 4.

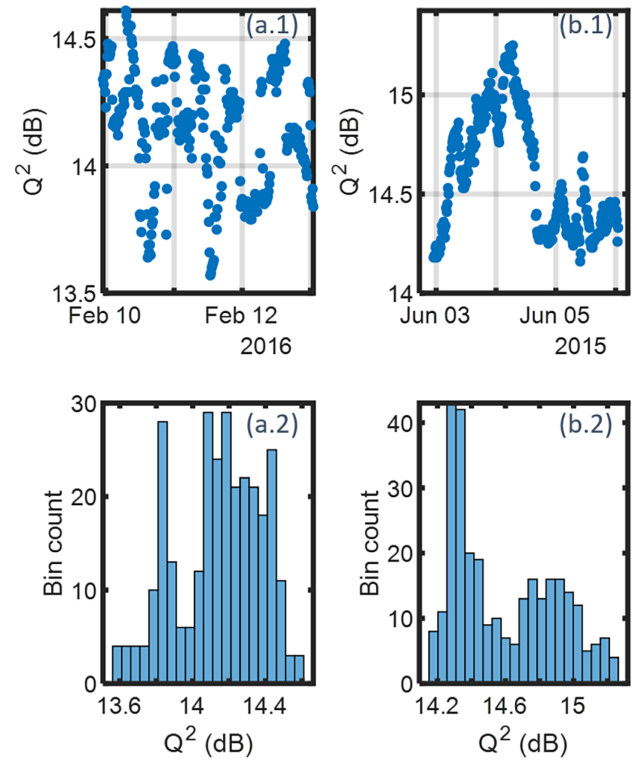


Fig. 4. Excerpts of QoT time series and corresponding histograms from lightpaths (a) #3151 and (b) #2522.

In Fig. 5(a), we plot the distribution of median MSD values for the 4000 lightpaths of the dataset. We observe a peak of the distribution close to 0.025 dB. Although not visible in the histogram, the highest measured values reach 0.15 dB, corresponding to the lightpaths plotted in Fig. 4. Further processing shows that the fluctuation range can be roughly estimated as six times the median MSD. In Fig. 5(b), the distribution of the maximum moving range shows that the issue of large QoT fluctuations is not isolated, with 20% of lightpaths showing fluctuations above 0.5 dB, and 4% above 1 dB.

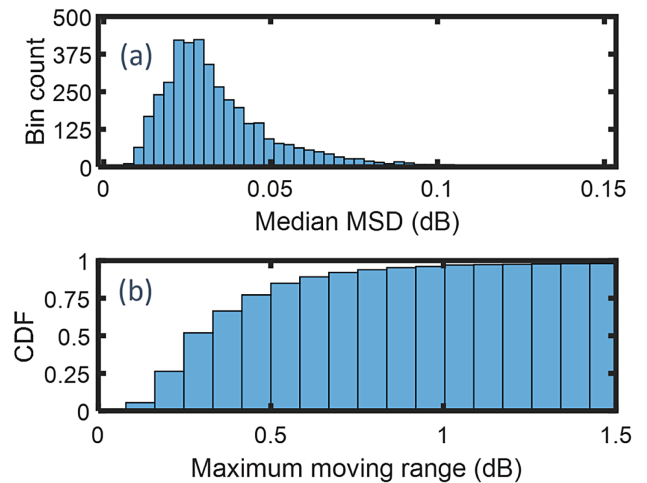


Fig. 5. (a) Histogram of the median MSD and (b) cumulated distribution function (CDF) of the maximum moving range for the 4000 lightpaths in the dataset.

3. IDENTIFICATION OF PDL AMPLITUDE AND ORIENTATION FROM QoT TIME SERIES

Since neither power fluctuations nor small distributed PDL elements can explain the QoT ranges and distributions observed in Section 2, we assume the presence of a single PDL anomaly, fully responsible for the observed fluctuations. Note that we can interpret this process as finding the characteristics of a single PDL element whose impact on QoT is similar to that of a more complex chain of PDL elements. In that regard, the approach is similar to the estimation of PDL at the coherent receiver from the CMA taps, as demonstrated in [17]. The equivalent model obtained may be helpful to build a simple data-driven model for any network lightpath.

In the following, we assume a PDL with fixed amplitude so that QoT fluctuations only stem from the variations of the PDL's orientation. The identification of the PDL element's parameters relies on the assumption that QoT time series are observed over a sufficient time period, allowing for the PDL orientations to reach both best and worst cases mentioned in Appendix A. Accordingly, the observed QoT extremes Q_{BC} and Q_{WC} should verify Eqs. (A21) and (A22). From Eq. (A22), we obtain

$$\beta = \frac{\max(Q^2)}{2Q_0^2 - \max(Q^2)}, \quad (1)$$

where, from Eq. (A21), Q_0^2 verifies

$$\mathcal{F}\left(\frac{Q_0^2 \max(Q^2)}{2Q_0^2 - \max(Q^2)}, Q_0^2\right) = \min(Q^2). \quad (2)$$

Here, Eq. (2) is a nonlinear equation that can be numerically solved. In practice, Eq. (2) has two solutions respectively corresponding to $\beta > 1$ and $\beta < 1$. By definition of the PDL magnitude β , we only retain the solution corresponding to $\beta < 1$ as valid, and the β value is deduced from the Q_0^2 value through Eq. (1). Once Q_0^2 and β values are identified, the θ value associated with each QoT sample Q_k^2 is determined solving the following nonlinear equation:

$$\mathcal{F}(Q_H^2(\theta), Q_V^2(\theta)) = Q_k^2, \quad (3)$$

where $Q_{H/V}^2(\theta)$ expressions are given by Eqs. (A14) and (A15). An inherent challenge of the resolution of the equation above is that the Q^2 is a $\pi/2$ -periodical and even function of θ , so that Eq. (3) has an infinite number of solutions. Since our goal is to test the consistency of the PDL model rather than properly measuring θ values, we assume in the following—for simplification—that θ may only take values between 0 and $\pi/4$.

We test the proposed identification scheme on the first 2000 samples of the QoT for the lightpath #3151, approximately corresponding to 20 days of data. The raw data is represented in Fig. 6 with its upper and lower envelopes, corresponding to the moving minimum and maximum over 24 h.

The extreme values for the 2000 QoT samples considered are $\min(Q^2[\text{dB}]) = 13.87$ dB and $\max(Q^2[\text{dB}]) = 14.7$ dB. With such values, the only valid solution of Eq. (2) is $Q_0^2[\text{dB}] = 16.30$ dB, which leads to $\beta[\text{dB}] = 2.68$ dB. Note that the value obtained for $Q_0^2[\text{dB}]$, i.e., the virtual Q^2

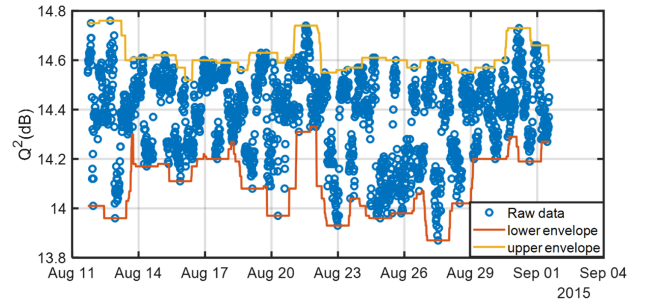


Fig. 6. Fitted QoT field data with lower and upper envelopes.

in absence of PDL, is consistent with a PM-QPSK lightpath of a few hundreds of km in length while the obtained PDL magnitude is clearly out of spec for any optical network device, but may be explained for instance by one defective node.

After solving Eq. (3) with θ between 0 and $\pi/4$, we plot the results in Fig. 7. As can be observed in Fig. 7(a), the samples reconstructed through the single-PDL model match the raw QoT samples after suitable identification of Q_0^2 and β . This is expected considering that the Q^2 is a bijective function of θ over $[0, \pi/4]$. As pointed out above, the θ values plotted in Fig. 7(b) are merely indicative due to the infinity of possible solutions of Eq. (3). Also, the fact that θ values explore all values from 0 to $\pi/4$ is a direct result of our hypothesis that the QoT is monitored over sufficient time to observe the best and worst cases. Considering for instance that the observed maximum is lower than the real maximum would result in a higher estimated PDL magnitude, counterbalanced by a lower range of variation for θ . Thus, we see that the mere observation of the Q^2 time series is enough to reinforce the hypothesis that the observed fluctuations are due to a PDL anomaly or equivalent, but the exact value of the PDL amplitude may only be determined after independent measurement of the orientation θ , for instance within the coherent receiver through the CMA taps [28]. Furthermore, the hypothesis of a PDL anomaly could be definitely confirmed by the joint observation of the QoT for both tributaries H and V . Indeed, as can be deduced from the

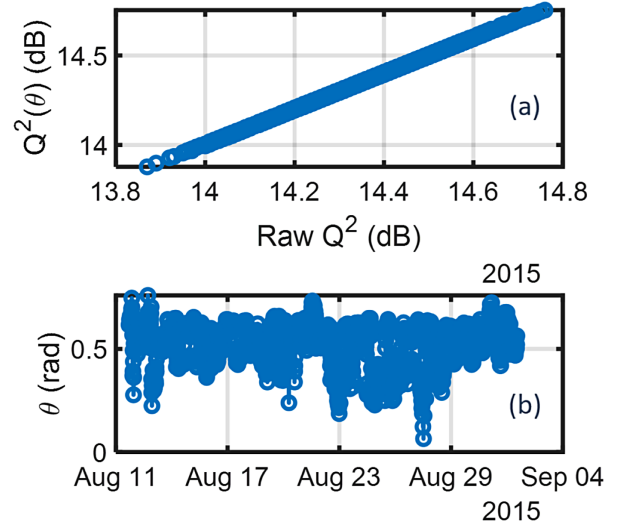


Fig. 7. (a) Q^2 reconstructed from θ versus the raw Q^2 value and (b) identified θ values versus time.

model derived in Appendix A, a time-dependent mismatch between Q_H^2 and Q_V^2 may only be explained by PDL.

4. NUMERICAL SIMULATIONS OF THE IMPACT OF PDL ON QOT FLUCTUATIONS IN AN OPTICAL NETWORK

In this section, we simulate the impact of distributed PDL on QoT fluctuations. In particular, we focus on the impact of a single PDL anomaly, i.e., a PDL element of abnormally large magnitude within a lightpath where the PDL is otherwise evenly distributed. Note that a PDL anomaly might also result from a concatenation of smaller but accidentally aligned PDL elements.

In Fig. 8, we represent an optical network with routing nodes connecting the various links, and lightpaths carrying data. A PDL anomaly—for instance from a defective node—affects some lightpaths while others experience typical distributed PDL. To model both scenarios, any network lightpath LP_k is represented as the concatenation of N_k elementary blocks in between the transmitter (Tx) and receiver (Rx) [9]. Each block schematically contains a span, a PDL element, and a PDL-free EDFA for optical amplification, all with known parameters representing the physical systems composing the lightpath. For simplicity, we neglect other impairments such as filtering, whose penalties can be independently accounted for [29]. We further assume a linear propagation regime, only considering the ASE noise added by the EDFAs.

We simulated a standard 100G PM-QPSK transmission as described in Fig. 9. The *SOP control* block is a rotation of angle θ_0 in the polarization plane for controlling the SOP at Tx output. It is coded as a pure rotation matrix in Jones space. Similarly, the repeated ROP blocks are rotations of angle θ_i , i from 1 to N , N standing for the total number of chained PDL blocks. These ROP blocks emulate the SOP fluctuations

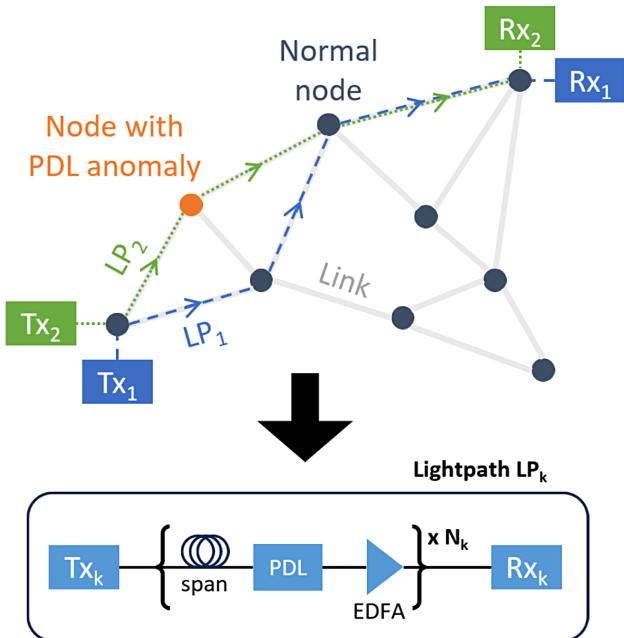


Fig. 8. Illustration of an optical network and its lightpaths, and their models as chains of PDL elements.

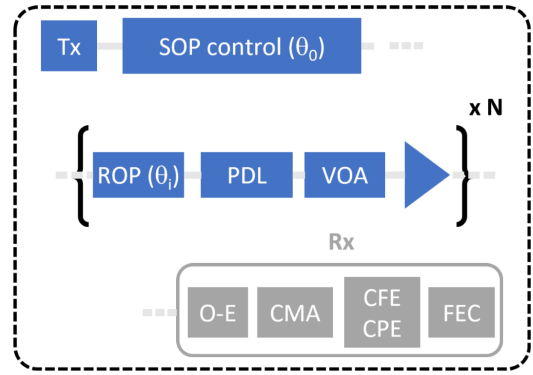


Fig. 9. Simulation setup.

between two consecutive PDL elements. Note that throughout the paper, we do not account for the ellipticity of the SOP as this parameter does not impact the QoT [18].

A. Impact of a Single PDL Element

Before simulating realistic lightpaths such as LP_1 and LP_2 in Fig. 8, we first consider a simplistic transmission with one inserted PDL element and compare the results with the analytical model derived in Appendix A. The PDL magnitude β [dB] is set at 2 dB. The gain of the EDFA is set to obtain Q_0^2 [dB] = 10 dB, the reference Q^2 without PDL. The orientation θ of the PDL element with regards to the tributaries is randomly set between $\pm\pi/2$ following a uniform distribution. The QoTs of the tributaries— Q_H^2 and Q_V^2 —are measured from the error counts after digital signal processing (DSP), obtained by comparing received and transmitted sequences. Finally, the Q^2 is derived from Q_H^2 and Q_V^2 using Eq. (A3). Note that in the dataset studied in Sections 2 and 3, the commercial transponders directly measure the Q^2 from an error count at FEC decoding [30].

We plot the simulation results in Fig. 10. Due to the PDL, Q^2 varies with θ_0 between two extremes. As expected from [18], the maximum Q^2 is achieved for $\theta_0 = \pi/4 + k\pi/2$ when $Q_H^2 = Q_V^2$, while the minimum is obtained for $\theta_0 = 0 + p\pi/2$

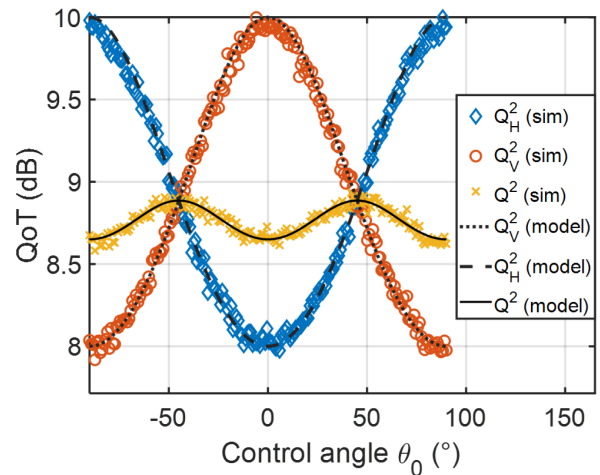


Fig. 10. Comparison between simulations (markers) and model (lines) for a back-to-back transmission with a single inserted PDL element, with Q_0 [dB] = 10 dB and β [dB] = 2 dB.

when $|Q_H^2 - Q_V^2|$ is maximal; k and p are integers. We observe that the simulated values are very well fitted by the direct estimation of Q^2 from the closed-form expression Eq. (A20) in Appendix A. The offset between model and simulations remains within ± 0.1 dB, corresponding to the QoT measurement uncertainty from the limited length of the test sequence [30].

B. Simulation of Realistic Optical Lightpaths

Next, we simulated large chains of PDL elements closely emulating the distributed PDL in arbitrary optical lightpaths. We first considered a chain of $N = 10$ PDL elements, first with the same 0.4 dB magnitude and random, uniformly distributed, θ_i angles renewed between each of the 100 iterations and $\theta_0 = 0$. As expected from the literature [9], the result in Fig. 11(a) shows that the arbitrary SOP fluctuations between the cascaded PDL elements tend to randomize $|Q_H^2 - Q_V^2|$. Since Q^2 directly depends on $|Q_H^2 - Q_V^2|$, as can be observed in Fig. 17, Q^2 may only slightly vary in time in such configurations.

We then emulate the presence of a single PDL anomaly by introducing a 4 dB-PDL among nine 0.4 dB-PDL elements. We test three positions for the anomaly: in the first, middle, and last positions in the PDL chain. The θ_i angles are randomly selected while θ_0 is scanned from $-\pi/2$ to $+\pi/2$. The result is plotted in Fig. 11(b). We observe that the impact of a given PDL element on the QoT imbalance between H and V significantly depends on its position within the lightpath. The impact is maximal when the PDL is close to the transmitter, and minimal when it is near the receiver. We also observe that at mid-distance, the impact is smaller but close to that of a PDL element near the transmitter. Finally, we notice that regardless of the position of the PDL anomaly, its impact largely dominates the variations of $|Q_H^2 - Q_V^2|$, so much so that the presence of one PDL anomaly tends to make an entire chain of PDL elements equivalent to one PDL element. Thus, in such cases, the Q^2 can be almost entirely controlled through the angle θ_0 at Tx despite random fluctuations of the polarization states between consecutive PDL elements. This is the rationale for the closed-loop PDL-mitigation scheme presented in Section 5.

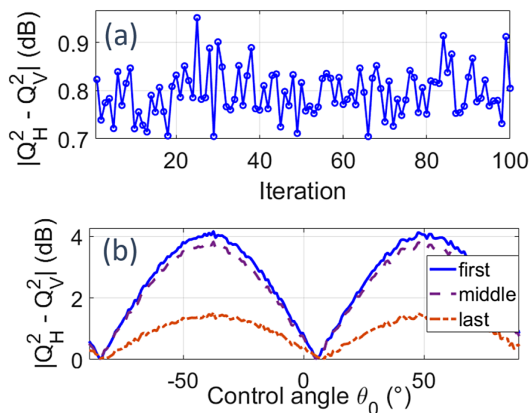


Fig. 11. Simulation results for $|Q_H^2 - Q_V^2|$ with $N = 10$ for (a) all PDL at 0.4 dB and (b) one PDL at 4 dB and nine at 0.4 dB.

5. CLOSED-LOOP AUTOMATION FOR MITIGATING PDL-INDUCED QOT FLUCTUATIONS

A. Working Principle

PDL-induced QoT fluctuations dominantly stem from the QoT mismatch between the two orthogonal tributaries created when a PDL component is located before an amplifier [18]. Thus, QoT fluctuations from PDL may not be fully mitigated at Rx through digital signal processing [31]. In parallel, substantial research showed that PDL and associated QoT fluctuations can be largely mitigated through spatiotemporal coding [32]. Our alternative solution tackles large PDL anomalies, where high QoT gains can be expected from the mitigation.

To mitigate PDL over an entire optical network, our solution consists of controlling the SOP at Tx for each lightpath (LP) through independent instances of the same control algorithm running in parallel. The solution can be implemented with a centralized control plane or by direct signaling between paired transponders, as illustrated in Fig. 12.

The SOP can be adjusted at Tx with minimal added complexity, either at digital-to-analog conversion, or through polarization control of the outgoing optical signal. For each LP, corrections of the SOP are occasionally triggered so that the measured Q_H^2 and Q_V^2 are maintained approximately equal, leading to a maximized QoT. Using $|\Delta Q^2| = |Q_H^2 - Q_V^2|$ as the error signal of the closed control loop is an essential aspect of our solution. Indeed, this guarantees that the automation is only triggered by an imbalance stemming from PDL, and not by other anomaly types that would also impact the Q^2 , but could not be mitigated through the SOP at Tx.

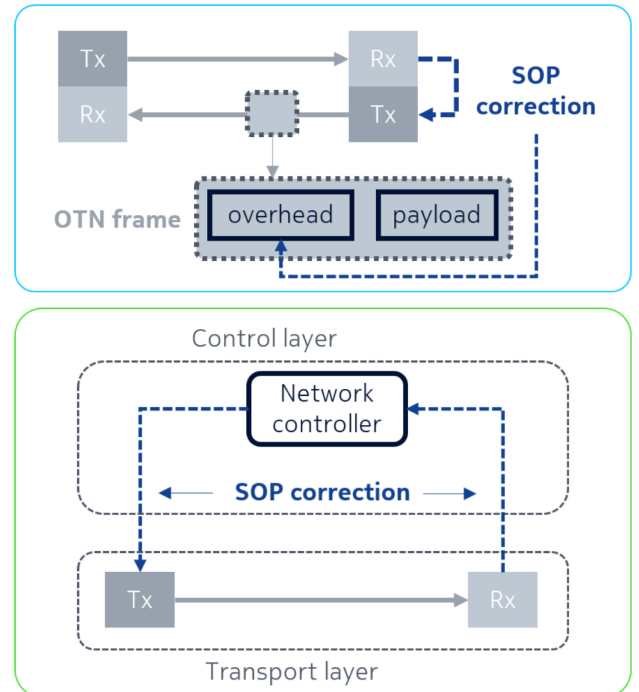


Fig. 12. Two possible solutions to close the SOP feedback loop: (a) by inserting the required SOP in the overhead of the OTN frame in the opposite direction and (b) through a centralized network controller.

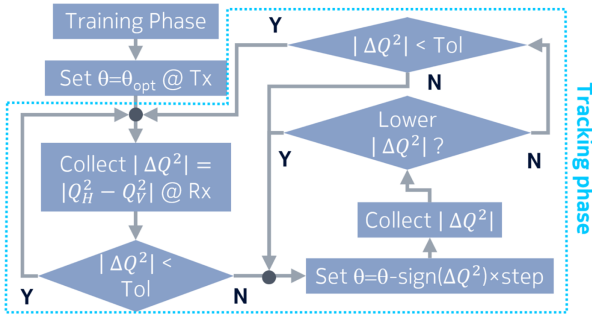


Fig. 13. Flowchart of the closed-loop SOP control loop for mitigating PDL-induced QoT fluctuations over a single lightpath.

As depicted in the flowchart of the control loop in Fig. 13, the automated cancellation of PDL-induced QoT fluctuations can be decomposed into two phases, training and tracking. The training is a calibration step meant to adjust every instance of the algorithm to the unique LP it controls. It consists of scanning the SOP at Tx output θ to characterize $\Delta Q^2 = Q_H^2 - Q_V^2 = f(\theta_{\text{PDL}})$ for the LP, as depicted in Fig. 14 where θ_{PDL} represents the input SOP for the PDL anomaly. The SOP $\theta = \theta_{\text{opt}}$ achieving $\Delta Q^2 = 0$ becomes the setpoint of the automation. During the tracking phase, the SOP is corrected to maintain $|\Delta Q^2|$ within tolerance, hence the QoT in the vicinity of its optimum value. Effective SOP correction is triggered when $|\Delta Q^2|$ exceeds the tolerance. The SOP correction is applied step-by-step, where the angular step is determined based on the training phase to ensure that the variation of $|\Delta Q^2|$ between steps is significantly higher than measurement noise. A correction cycle ends when $|\Delta Q^2|$ cannot be further reduced and is back within bounds. The algorithm is notably designed to function even if the training curve collected at startup becomes obsolete due to major SOP drifts or network reconfigurations. In such rare events, the control angle will be continuously modified until $|\Delta Q^2|$ is reduced and eventually returns within bounds.

B. Experiment

Our experimental testbed is based on an offline back-to-back transmission of a typical 100G PM-QPSK where we inserted a SOP controller, a SOP scrambler, and a PDL element between the Tx and the VOA. It corresponds to the representation in

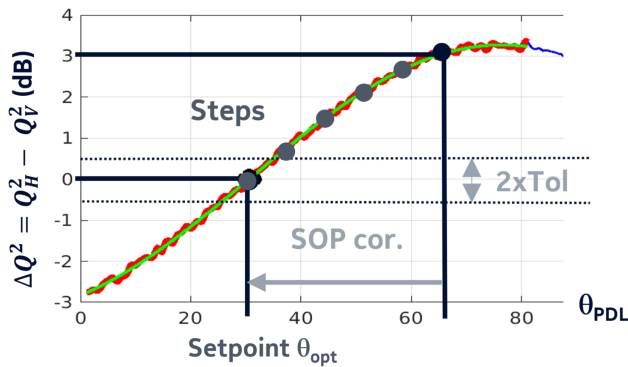


Fig. 14. Example of an experimental calibration curve obtained during the training phase to determine SOP setpoint θ_0 and convergence parameters.

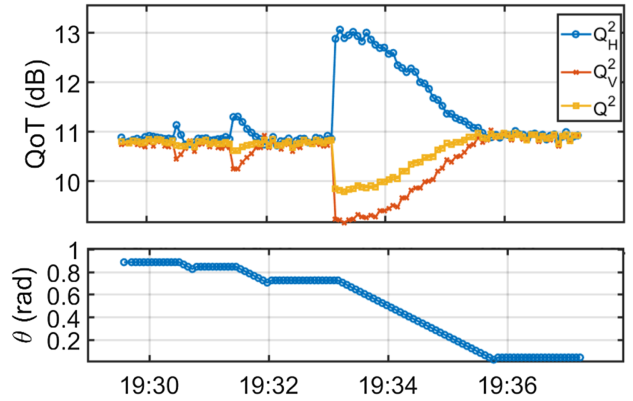


Fig. 15. PDL mitigation experiment with large and instantaneous perturbations of the SOP.

Fig. 9 with $N = 1$. The PDL is set at approximately 5 dB. In a first experiment, we test the robustness of the automation under large and fast SOP variations. As explained in the previous section, we start by the training phase where we measure ΔQ^2 as a function of the control angle θ applied through the SOP controller. Based on the collected data, we set the angular step to 0.02 rad. After training, we impose increasingly large step-like variations of the SOP at the input of the PDL through manual control of the SOP scrambler. The result is plotted in Fig. 15.

For each of the three step-like perturbations, the solution manages to recover the initial point corresponding to an optimal Q^2 around 11 dB. As expected, the larger the SOP perturbation, the longer it takes for the step-by-step mitigation to converge. This illustrates the necessary limitation from the response time of the closed loop when it comes to counteracting fast and large SOP variations. To test the automation with SOP perturbations closer to the typical rates observed in deployed optical networks, we set in a second experiment the SOP scrambler so that it continuously modifies the SOP on the Poincaré sphere following Rayleigh statistics, which best reproduces natural SOP fluctuations in optical fibers [10]. We plot the result in Fig. 16 obtained after overnight measurements. With realistic SOP fluctuations, our solution manages to maintain the Q^2 within ± 0.2 dB despite the large PDL anomaly in the experimental setup.

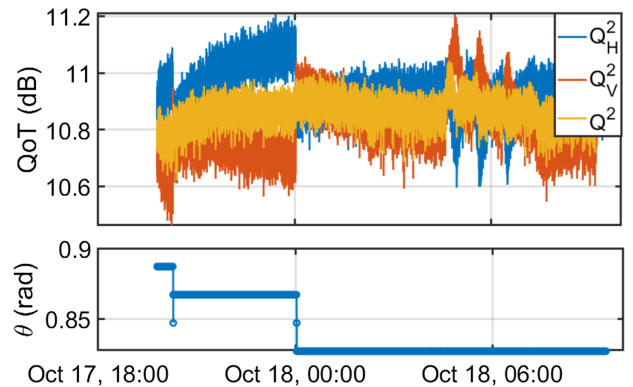


Fig. 16. PDL mitigation experiment with continuous SOP perturbations from Rayleigh statistics.

6. CONCLUSION

With a numerical method specifically designed to characterize QoT fluctuations in optical networks, we investigated their statistical distribution on the field, leveraging public monitoring data from a large backbone. We observed that 20% of lightpaths exhibit large fluctuation levels, over 0.5 dB in range, and that such fluctuations may hardly be explained by power fluctuations or typical distributed PDL. Instead, after thorough investigation, these QoT fluctuations proved largely consistent with the presence of large PDL anomalies within the monitored network. From the analytical model we derived, definite proof of the PDL origin of these field-observed QoT fluctuations would require simultaneous monitoring of the QoTs from both polarization tributaries. Meanwhile, to specifically address such potential PDL anomalies and their negative impact on the quality of service, we proposed a closed-loop automation that effectively stabilizes the QoT of an experimental lightpath within ± 0.2 dB despite 5 dB of PDL and realistic SOP fluctuations.

APPENDIX A: SOP-DEPENDENT QOT PENALTY INDUCED BY A SINGLE PDL ELEMENT

One specific impact of PDL in coherent, PM optical transmissions is a QoT imbalance between H and V tributaries, characterized by $Q_H^2 \neq Q_V^2$. This imbalance is random and time-dependent, as directly depending on the SOP variations between PDL elements. In the following, we present an analytical model describing how a PDL element leads to fluctuations of Q^2 , the compound QoT of the PM signal formed by both tributaries.

The QoTs of the H and V tributaries Q_H^2 and Q_V^2 derive from their respective pre-FEC BERs— BER_H and BER_V —as [24]

$$Q_{H,V}^2 = 2[\text{erfc}^{-1}(2BER_{H,V})]^2. \quad (\text{A1})$$

In a commercial transponder, the Q^2 is measured from an error count provided by the FEC decoder [30]. Thus, because of the interleaving of both tributaries within a FEC frame, the measured Q^2 derives from the average of BER_H and BER_V [18]:

$$Q^2 = 2[\text{erfc}^{-1}(BER_H + BER_V)]^2. \quad (\text{A2})$$

From Eqs. (A1) and (A2), we obtain

$$Q^2 = 2 \left\{ \text{erfc}^{-1} \left[\frac{1}{2} \text{erfc} \left(\sqrt{\frac{Q_H^2}{2}} \right) + \frac{1}{2} \text{erfc} \left(\sqrt{\frac{Q_V^2}{2}} \right) \right] \right\}^2. \quad (\text{A3})$$

For compactness, we define the function $\mathcal{F}$ so that

$$Q^2 = \mathcal{F}(Q_H^2, Q_V^2). \quad (\text{A4})$$

To illustrate how a QoT imbalance between the two tributaries impacts the overall QoT, let us first write Q_H^2 and Q_V^2 in dB as functions of their average $Q_{\text{avg}}^2[\text{dB}] = (Q_H^2[\text{dB}] + Q_V^2[\text{dB}])/2$ and difference $\Delta Q^2[\text{dB}] = Q_H^2[\text{dB}] - Q_V^2[\text{dB}]$:

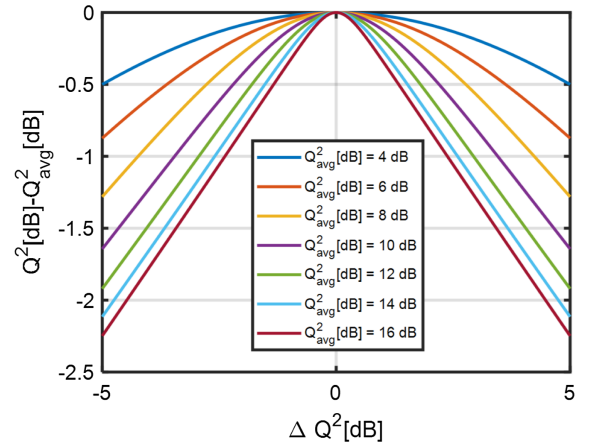


Fig. 17. $Q^2[\text{dB}]$ as a function of $\Delta Q^2[\text{dB}]$ for various values of the average QoT, $Q_{\text{avg}}^2[\text{dB}]$.

$$Q_{H,V}^2[\text{dB}] = Q_{\text{avg}}^2[\text{dB}] \pm \frac{\Delta Q^2[\text{dB}]}{2}. \quad (\text{A5})$$

In Fig. 17, we plot the deviation of Q^2 from Q_{avg}^2 as a function of $\Delta Q^2[\text{dB}]$ for growing values of Q_{avg}^2 . The main observation is that the higher the Q_{avg}^2 , the more the Q^2 value—deriving from Eq. (A5)—deviates from Q_{avg}^2 . This for instance means that a PDL element leading to an imbalance between Q_H^2 and Q_V^2 will lead to larger fluctuations of Q^2 for shorter lightpaths with strong signal quality than for longer ones.

In [18], Duthel *et al.* studied the impact of a single PDL element on PM coherent transmissions. The authors specifically studied the two extreme values taken by the QoT depending—in Jones space—on the angle θ —between the PDL axis and the orthogonal, linearly polarized Tx lasers of both H and V tributaries. This is illustrated in Fig. 18, where we introduce the orthonormal basis $(\mathbf{u}_H, \mathbf{u}_V)$ so that $\mathbf{s}_H = s_H \mathbf{u}_H$ and $\mathbf{s}_V = s_V \mathbf{u}_V$ are the Jones vectors associated with the H and V tributaries at the transmitter's output. We also introduce the orthonormal basis $(\mathbf{u}_{\text{PDL}}, \mathbf{v}_{\text{PDL}})$ formed with the two principal axes of the PDL element.

Similar to [18], we quantify the PDL through the extra power attenuation β along the direction $\mathbf{u}_{\text{PDL}}$ compared to $\mathbf{v}_{\text{PDL}}$. We now write $\mathbf{s}_H$ and $\mathbf{s}_V$ in $(\mathbf{u}_{\text{PDL}}, \mathbf{v}_{\text{PDL}})$, introducing the total power at emission $p_0 = s_H^2 + s_V^2$ and assuming $s_H^2 = s_V^2$:

$$\begin{cases} \mathbf{s}_H = \sqrt{\frac{p_0}{2}} [\cos(\theta) \mathbf{u}_{\text{PDL}} - \sin(\theta) \mathbf{v}_{\text{PDL}}], \\ \mathbf{s}_V = \sqrt{\frac{p_0}{2}} [\sin(\theta) \mathbf{u}_{\text{PDL}} + \cos(\theta) \mathbf{v}_{\text{PDL}}]. \end{cases} \quad (\text{A6})$$

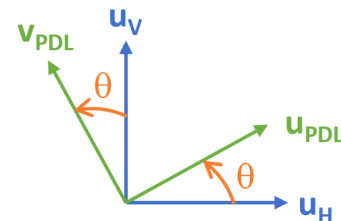


Fig. 18. Representation of the H and V tributaries and principal axes of a single PDL element in Jones polarization space.

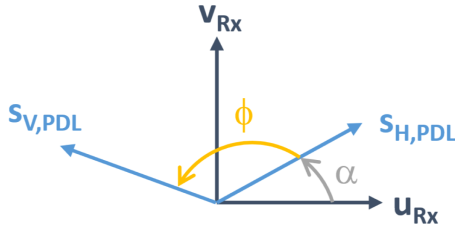


Fig. 19. Representation of the H and V tributaries before reception and principal axes of the receiver in Jones polarization space.

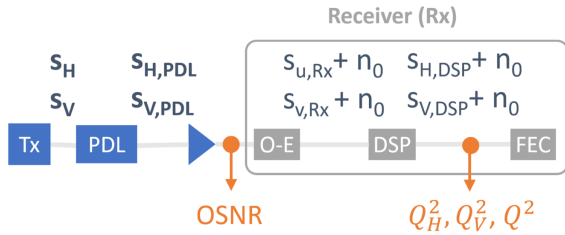


Fig. 20. Back-to-back transmission with an inserted PDL element.

After the PDL element, because of the extra power attenuation along $\mathbf{u}_{\text{PDL}}$, the two tributaries become

$$\begin{cases} \mathbf{s}_{H,\text{PDL}} = \sqrt{\frac{p_0}{2}} [\sqrt{\beta} \cos(\theta) \mathbf{u}_{\text{PDL}} - \sin(\theta) \mathbf{v}_{\text{PDL}}], \\ \mathbf{s}_{V,\text{PDL}} = \sqrt{\frac{p_0}{2}} [\sqrt{\beta} \sin(\theta) \mathbf{u}_{\text{PDL}} + \cos(\theta) \mathbf{v}_{\text{PDL}}]. \end{cases} \quad (\text{A7})$$

Using the scalar product of $\mathbf{s}_{H,\text{PDL}}$ and $\mathbf{s}_{V,\text{PDL}}$, we derive from Eq. (A7) the following equation:

$$\mathbf{s}_{H,\text{PDL}} \cdot \mathbf{s}_{V,\text{PDL}} = \sqrt{p_{H,\text{PDL}} p_{V,\text{PDL}}} \cos(\phi), \quad (\text{A8})$$

where ϕ is the angle between both tributaries in Jones space, and $p_{H,\text{PDL}}$ and $p_{V,\text{PDL}}$ are their respective powers, verifying

$$\begin{cases} p_{H,\text{PDL}} = \|\mathbf{s}_{H,\text{PDL}}\|^2 = \frac{p_0}{2} [\beta \cos^2(\theta) + \sin^2(\theta)], \\ p_{V,\text{PDL}} = \|\mathbf{s}_{V,\text{PDL}}\|^2 = \frac{p_0}{2} [\beta \sin^2(\theta) + \cos^2(\theta)]. \end{cases} \quad (\text{A9})$$

At reception, both tributaries are projected on the orthonormal basis $(\mathbf{u}_{\text{Rx}}, \mathbf{v}_{\text{Rx}})$, illustrated in Fig. 19 where α is an arbitrary angle. Thus, the received signals after coherent mixing $s_{u,\text{Rx}}$ and $s_{v,\text{Rx}}$ are expressed as

$$\begin{cases} s_{u,\text{Rx}} = \sqrt{p_{H,\text{PDL}}} \cos(\alpha) + \sqrt{p_{V,\text{PDL}}} \cos(\alpha + \phi), \\ s_{v,\text{Rx}} = \sqrt{p_{H,\text{PDL}}} \sin(\alpha) + \sqrt{p_{V,\text{PDL}}} \sin(\alpha + \phi). \end{cases} \quad (\text{A10})$$

After digital signal processing, assuming zero-forcing, the demultiplexed signals before decision can be written:

$$\begin{cases} s_{H,\text{DSP}} = \sin(\alpha + \phi) s_{u,\text{Rx}} - \cos(\alpha + \phi) s_{v,\text{Rx}}, \\ s_{V,\text{DSP}} = \sin(\alpha) s_{u,\text{Rx}} - \cos(\alpha) s_{v,\text{Rx}}. \end{cases} \quad (\text{A11})$$

After simplification, we obtain

$$s_{H/V,\text{DSP}} = \sqrt{p_{H/V,\text{PDL}}} \sin(\phi). \quad (\text{A12})$$

This leads to the expressions of the respective powers for both tributaries at decision, as illustrated in Fig. 20:

$$\begin{cases} p_{H,\text{DSP}} = \frac{p_0}{2} [\beta \cos^2(\theta) + \sin^2(\theta)] \sin^2(\phi), \\ p_{V,\text{DSP}} = \frac{p_0}{2} [\beta \sin^2(\theta) + \cos^2(\theta)] \sin^2(\phi). \end{cases} \quad (\text{A13})$$

Assuming that the total transmission noise n_0 is not polarized, the associated noise power $\sigma_0^2 = \text{Var}(n_0)$ prior to decision—not impacted by the demultiplexing—is the same for both tributaries. Consequently, Q_H^2 and Q_V^2 can finally be written as functions of the PDL ratio β and its orientation θ :

$$Q_{H/V}^2 = Q_0^2 \frac{1+\beta}{2} \left[1 \mp \frac{1-\beta}{1+\beta} \cos(2\theta) \right] \sin^2(\phi), \quad (\text{A14})$$

where $Q_0^2 = p_0/\sigma_0^2$ is the QoT in absence of PDL ($\beta = 1$) and $\sin(\phi)$ derives from Eqs. (A7) and (A8):

$$\sin^2(\phi) = 1 - \cos^2(\phi) = \left[1 + \frac{(\beta-1)^2}{4\beta} \sin^2(2\theta) \right]^{-1}. \quad (\text{A15})$$

From Eq. (A14), we verify that Q_H^2 and Q_V^2 are π -periodical functions of θ so that the study can be limited to $[-\pi/2, \pi/2]$. We further verify that the expressions for Q_H^2 and Q_V^2 obtained when $\theta = 0 \pm \pi/2$ and $\theta = \pm\pi/4$ match the ones derived by Duthel *et al.* When $\theta = 0 \pm \pi/2$, the tributaries remain orthogonal after the PDL element so that $\cos(\phi) = 0$ and $\cos(2\theta) = \pm 1$, and the offset of power attenuation between the two tributaries is maximal leading to the highest QoT imbalance:

$$\max(|\Delta Q^2[\text{dB}]|) = |\Delta Q^2[\text{dB}]|(\theta = 0 \pm \pi/2) = -\beta[\text{dB}]. \quad (\text{A16})$$

In contrast, when $\theta = \pm\pi/4$, $\sin(\phi) = (\beta+1)^2/4\beta$, and $\cos(2\theta) = 0$, we verify

$$Q_H^2(\theta = \pm\pi/4) = Q_V^2(\theta = \pm\pi/4) = \frac{2\beta}{1+\beta} Q_0^2. \quad (\text{A17})$$

For further insight, we can decompose the QoT penalty due to the PDL element into three contributions: $\mathcal{P}_0$, $\mathcal{P}_{\Delta\text{att}}$, and $\mathcal{P}_{\text{depol}}$. The first term is a direct result of the overall attenuation of the PDL element for the PM signal. The second term is the result of the attenuation offset between the tributaries. Finally, the third term results from the depolarization characterized by the non-orthogonality of $\mathbf{s}_{H,\text{PDL}}$ and $\mathbf{s}_{V,\text{PDL}}$, whose correction by the DSP is equivalent to a power attenuation of $\sin^2(\phi)$ for each tributary. Thus, we can write

$$Q_{H/V}^2 = \mathcal{P}_{\text{PDL},H/V} Q_0^2 = \mathcal{P}_0 \mathcal{P}_{\Delta\text{att},H/V} \mathcal{P}_{\text{depol}} Q_0^2, \quad (\text{A18})$$

where

$$\begin{cases} \mathcal{P}_0 = \frac{1+\beta}{2}, \\ \mathcal{P}_{\Delta\text{att},H/V} = 1 \mp \frac{1-\beta}{1+\beta} \cos(2\theta), \\ \mathcal{P}_{\text{depol}} = \left[1 + \frac{(\beta-1)^2}{4\beta} \sin^2(2\theta) \right]^{-1}. \end{cases} \quad (\text{A19})$$

According to Eq. (A4), the PDL-induced penalty $\mathcal{P}_{\text{PDL}}$ on Q^2 verifies

$$Q^2 = \mathcal{P}_{\text{PDL}} Q_0^2 = \mathcal{F}(\mathcal{P}_{\text{PDL},H} Q_0^2, \mathcal{P}_{\text{PDL},V} Q_0^2). \quad (\text{A20})$$

With $\mathcal{P}_{\text{PDL},H}(\theta = 0) = \beta$ and $\mathcal{P}_{\text{PDL},V}(\theta = 0) = 1$, $\theta = 0$ corresponds to the worst-case of PDL penalty, verifying

$$Q_{\text{WC}}^2 = \mathcal{P}_{\text{PDL},\text{WC}} Q_0^2 = \mathcal{F}(\beta Q_0^2, Q_0^2). \quad (\text{A21})$$

In opposition, with $\mathcal{P}_{\text{PDL},H}(\theta = \pm\pi/4) = \mathcal{P}_{\text{PDL},V}(\theta = \pm\pi/4) = 2\beta/(1+\beta)$, $\theta = \pm\pi/4$ corresponds to the best-case PDL penalty, verifying

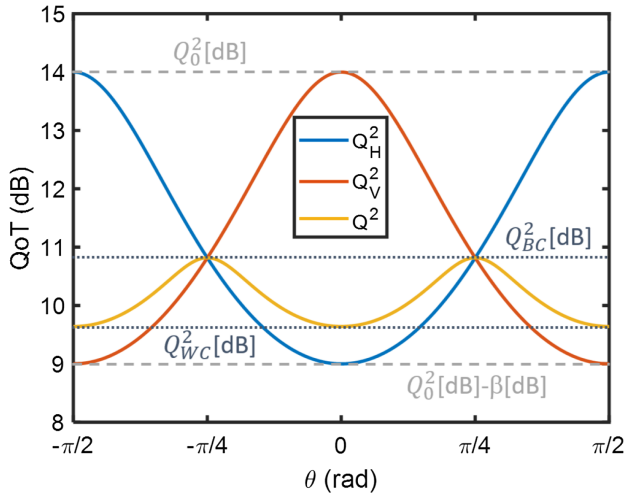


Fig. 21. Q_H^2 [dB], Q_V^2 [dB], and Q^2 [dB] as functions of θ for Q_0^2 [dB] = 14 dB and β [dB] = 5 dB.

$$Q_{BC}^2 = \mathcal{P}_{PDL,BC} Q_0^2 = \frac{2\beta}{1+\beta} Q_0^2. \quad (\text{A22})$$

In Fig. 21, we plot the variations of Q_H^2 [dB], Q_V^2 [dB], and Q^2 [dB] as functions of θ for Q_0^2 [dB] = 14 dB and β [dB] = 5 dB. For $\theta = \pm\pi/2$, the situation is reversed compared to $\theta = 0$ but leads to the same worst-case PDL penalty.

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